Bahria University, Islamabad

Department of Software Engineering

Artificial Intelligence Lab (Spring-2026)

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Lab Journal: 02

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Task No:	Task Wise Marks		Documentation Marks		Total Marks (20)
	Assigned	Obtained	Assigned	Obtained	
1	3		5		
2	3				
3	3				
4	3				
5	3				

Comments:

Signature

Lab No: 02 - Collections

Introduction

Introduction to Collections in Python

Tools Used

- Python
- VS Code

Task 1:

Dr. John Wesley has a spreadsheet containing a list of student's ID, marks, class and name.

Your task is to help Dr. Wesley calculate the average marks of the students.

Average = Sum of all marks/Total Students

Note:

1. Columns can be in any order. IDs, marks, class, and name can be written in any order in the spreadsheet.

2. Columns names are ID, MARKS, CLASS and NAME. (The spelling and cast type of these names won't change).

Input Format:

The first line contains an integer N, the total number of students The second line contains the names of the columns in any order.

The next N lines contains the MARKS, IDs, NAME and class, under their respective column names.

Constraints:

$0 < N <= 100$

Output Format: Print the average marks of the list corrected to 2 decimal places.

CODE:

```python
def calculate_average(N, columns, rows):
    if N == 0 or "MARKS" not in columns:
        return 0.00

    marks_index = columns.index("MARKS")
    total_marks = 0.0
    valid_count = 0

    for row in rows:
        if len(row) > marks_index:
            try:
                total_marks += float(row[marks_index])
                valid_count += 1
            except ValueError:
                continue

    if valid_count == 0:
        return 0.00

    return total_marks / valid_count


def main():
    try:
        N = int(input("Enter total number of students: ").strip())
        columns = input(
            "Enter column names separated by space: ").strip().split()

        rows = []
        print("Enter student records (one per line):")
        for i in range(N):
            row = input(f"Student {i+1}: ").strip().split()
            rows.append(row)

        average = calculate_average(N, columns, rows)

        print(f"\nAverage Marks: {average:.2f}")

    except Exception:
        print("Invalid input. Average Marks: 0.00")


if __name__ == "__main__":
    main()
```

Screenshot:

```
Talhas-MacBook-Air:Lab 02 talha$ python3 Exercise.py
Enter total number of students: 3
Enter column names separated by space: ID NAME MARKS CLASS
Enter student records (one per line):
Student 1: 1 Talha 85 10
Student 2: 2 Arham 78 10
Student 3: 3 Qayyum 92 9

Average Marks: 85.00
Talhas-MacBook-Air:Lab 02 talha$
```

Figure 1. Output

Task 2:

In this challenge, the user enters a string and a sub-string. You have to print the number of times that the sub-string occurs in the given string. String traversal will take place from left to right, not from right to left.

Note: String letters are case-sensitive.

Input Format:

The first line of input contains the original string. The next line contains the sub-string.

Constraints:

1<= len(string) <= 200

Output Format: Output the integer number indicating the total number of occurrences of the sub-string in the original string.

Code:

```python
def count_substring_occurrences(string, sub_string):
    count = 0
    sub_len = len(sub_string)
    str_len = len(string)

    for i in range(str_len - sub_len + 1):
        if string[i:i + sub_len] == sub_string:
            count += 1

    return count
```

```python
def main():
    try:
        string = input("Enter the original string: ").strip()
        sub_string = input("Enter the sub-string: ").strip()

        if not (1 <= len(string) <= 200):
            print("String length must be between 1 and 200.")
            return

        result = count_substring_occurrences(string, sub_string)

        print(f"\nNumber of occurrences: {result}")

    except Exception:
        print("Invalid input.")


if __name__ == "__main__":
    main()
```

Screenshot:

```
Talhas-MacBook-Air:Lab 02 talha$ python3 Task.py
Enter the original string: ABABABA
Enter the sub-string: ABA

Number of occurrences: 3
Talhas-MacBook-Air:Lab 02 talha$
```

Figure 2. Output

Task 3: Count the occurrences of each character in the input_str.

Input:

Take an input from user named input_str.

Output:

Print each character and its count.

Sample:

IN: aabcd

OUT: a:2, b:1, c:1, d:1

Code:

```python
def count_characters(input_str):
    char_count = {}
    order = []

    for ch in input_str:
        if ch not in char_count:
            char_count[ch] = 1
            order.append(ch)
        else:
            char_count[ch] += 1

    result = ", ".join(f"{ch}:{char_count[ch]}" for ch in order)
    return result


def main():
    input_str = input("Enter the string: ").strip()

    output = count_characters(input_str)
    print(f"\n{output}")


if __name__ == "__main__":
    main()
```

Screenshot:

```
Talhas-MacBook-Air:Lab 02 talha$ python3 Task_2.py
Enter the string: aabcd

a:2, b:1, c:1, d:1
Talhas-MacBook-Air:Lab 02 talha$
```

Figure 3. Output

Conclusion:

In completing these tasks, I learned how to manipulate strings and work with data in Python effectively. I practiced counting elements, handling dynamic input, and ensuring my solutions were both robust and efficient. I also understood the importance of preserving order and formatting output correctly. Overall, these exercises helped me strengthen my problem-solving skills and gave me confidence in applying Python to real-world scenarios.